

Solution Requirements

Date	6 Jun 2026
Team ID	
Project Name	OptiCrop
Maximum Marks	4 Marks

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	Environmental Parameter Form Input	System must provide interactive sliding or numerical entries for seven attributes: Nitrogen (N), Phosphorous (P), Potassium (K), Temperature, Humidity, Soil pH, and Rainfall with active value boundary validations.
FR-2	Machine Learning Crop Prediction	System must invoke a trained Random Forest Classification module to process incoming user criteria strings and produce matching optimal crop labels.
FR-3	Target Crop Evaluation Mapping	System must extract dynamic classifier prediction probabilities to evaluate compatibility scores for specific user-selected crop variations.
FR-4	Macro Research Matrix Output	System must generate aggregate historical mean parameter summaries grouped directly by plant profiles for strategic regional policymakers.

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

FR No.	Non-functional Requirement	Description
NFR-1	Usability	The web system layout must remain intuitive, distraction-free, and lightweight, ensuring seamless operational use by remote smallholder farming entities.
NFR-2	Security	Client parameter limits are verified at submission to ensure execution protection from calculation injections or memory buffer script overrides.
NFR-3	Reliability	The core predictive library operations must remain stateless, maintaining reliable 99.9% prediction accuracy across repeated standard test sequences.

NFR-4	Performance	
FR No.	Non-functional Requirement	Description
		The system model calculation loop and diagnostic results viewport rendering must update fully in under 2.0 seconds from trigger input receipt.
NFR-5	Availability	The computational engine pipeline is built for high continuous up-time availability deployment blocks across major agricultural planting seasons.
NFR-6	Scalability	The application's decoupled architectural layout allows scaling instances seamlessly to handle bulk processing tasks during peak community use.